

Michael Smith
Linux admin
Continue set up
Centos

```
ms3213@localhost:~$ yum repolist
repo id                                repo name
appstream                              CentOS Stream 10 - AppStream
baseos                                 CentOS Stream 10 - BaseOS
extras-common                          CentOS Stream 10 - Extras packages
ms3213@localhost:~$
```

1.

```
Last metadata expiration check: 0:00:01 ago on Mon 17 Nov 2025 12:43:25 AM EST.
Repo-id                                : appstream
Repo-name                              : CentOS Stream 10 - AppStream
Repo-revision                          : 10-stream
Repo-distro-tags                       : lcp:/o:centos-stream:centos-stream:101: , , 0, 1, C, 0, S, S, a, e, e, m, n, r, t, t
Repo-updated                           : Tue 11 Nov 2025 01:00:59 PM EST
Repo-pkgs                              : 14,434
Repo-available-pkgs                    : 14,434
Repo-size                              : 23 G
Repo-metalink                          : https://mirrors.centos.org/metalink?repo=centos-appstream-10-stream&arch=x86_64&protocol=https,h
  Updated                              : Mon 17 Nov 2025 12:43:20 AM EST
Repo-baseurl                           : http://ftp-chi.osuosl.org/pub/centos-stream/10-stream/AppStream/x86_64/os/ (42 more)
Repo-expire                            : 21,600 second(s) (last: Mon 17 Nov 2025 12:43:20 AM EST)
Repo-filename                          : /etc/yum.repos.d/centos.repo

Repo-id                                : baseos
Repo-name                              : CentOS Stream 10 - BaseOS
Repo-revision                          : 10-stream
Repo-distro-tags                       : lcp:/o:centos-stream:centos-stream:101: , , 0, 1, C, 0, S, S, a, e, e, m, n, r, t, t
Repo-updated                           : Tue 11 Nov 2025 12:58:48 PM EST
Repo-pkgs                              : 3,824
Repo-available-pkgs                    : 3,824
Repo-size                              : 6.2 G
Repo-metalink                          : https://mirrors.centos.org/metalink?repo=centos-baseos-10-stream&arch=x86_64&protocol=https,http
  Updated                              : Mon 17 Nov 2025 12:43:10 AM EST
Repo-baseurl                           : http://ftp-chi.osuosl.org/pub/centos-stream/10-stream/BaseOS/x86_64/os/ (42 more)
Repo-expire                            : 21,600 second(s) (last: Mon 17 Nov 2025 12:43:10 AM EST)
Repo-filename                          : /etc/yum.repos.d/centos.repo

Repo-id                                : extras-common
Repo-name                              : CentOS Stream 10 - Extras packages
Repo-revision                          : 1762251386
Repo-updated                           : Tue 04 Nov 2025 05:16:26 AM EST
Repo-pkgs                              : 50
Repo-available-pkgs                    : 50
Repo-size                              : 497 k
Repo-metalink                          : https://mirrors.centos.org/metalink?repo=centos-extras-sig-extras-common-10-stream&arch=x86_64&p
  Updated                              : Mon 17 Nov 2025 12:43:25 AM EST
Repo-baseurl                           : http://ftp-chi.osuosl.org/pub/centos-stream/SIGs/10-stream/extras/x86_64/extras-common/ (42 more)
Repo-expire                            : 21,600 second(s) (last: Mon 17 Nov 2025 12:43:25 AM EST)
Repo-filename                          : /etc/yum.repos.d/centos-addons.repo
Total packages: 18,308

ms3213@localhost:~$ yum -v repolist
Loaded plugins: builddep, changelog, config-manager, copr, debuginfo-install, dnf-plugins-core,
ifypackagekit, playground, repoclosure, repodiff, repograph, repomanage, repository-plugin
YUM version: 4.20.0
cachedir: /var/tmp/dnf-ms3213-iwvt_0h0
CentOS Stream 10 - BaseOS
CentOS Stream 10 - AppStream
```

<https://www.cyberciti.biz/faq/centos-fedora-redhat-yum-repolist-command-to-list-package-repositories/>

The commands **yum -v repolist**, **yum -v repolist | less**, and **yum repolist** can be used to list repository information. We see repositories here, such as baseos and appstream.

CentOS does not use PPAs, but repositories can be installed using the **yum install** command.
<https://unix.stackexchange.com/questions/593388/installing-packages-on-centos-7>

```
ms3213@localhost:~$ sudo yum install tmux
```

```
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:
```

- #1) Respect the privacy of others.
- #2) Think before you type.
- #3) With great power comes great responsibility.

```
For security reasons, the password you type will not be visible.
```

```
[sudo] password for ms3213:
```

2. ~~Setup tmux~~

```
Installing:
tmux                                     x86_64                               3.3a-13.2023
Transaction Summary
-----
Install 1 Package

Total download size: 518 k
Installed size: 1.2 M
Is this ok [y/N]: y
Downloading Packages:
tmux-3.3a-13.20230918gitb202a2f.el10.x86_64.rpm
-----
Total
CentOS Stream 10 - BaseOS
Importing GPG key 0x8483C65D:
  Userid      : "CentOS (CentOS Official Signing Key) <security@centos.org>"
  Fingerprint: 99DB 70FA E1D7 CE22 7FB6 4882 05B5 55B3 8483 C65D
  From        : /etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial-SHA256
Is this ok [y/N]: y
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 
  Installing     : tmux-3.3a-13.20230918gitb202a2f.el10.x86_64
  Running scriptlet: tmux-3.3a-13.20230918gitb202a2f.el10.x86_64

Installed:
  tmux-3.3a-13.20230918gitb202a2f.el10.x86_64

Complete!
```

Using what we learned before, I tried “**yum install tmux**”, which then asked for supervisor privileges. So then I **ran sudo yum install tmux** to elevate permissions and allow installation. The two pictures above show the command to install, and the second one shows what it looks like during installation. When run, it looks like the picture below. This shows that a new terminal session is open within the terminal, as tmux allows for multiple terminals at once.

```
[0] 0:bash* "localhost"
```

To exit, click the keys Ctrl and b, then release and press d

```
ms3213@localhost:~$ sudo yum install emacs-nox
[sudol password for ms3213:
Last metadata expiration check: 3:20:28 ago on Mon 17 Nov 2025 03:02:24 AM EST.
Dependencies resolved.
=====
Package                        Architecture                Version
=====
Installing:
  emacs-nox                    x86_64                      1:29.4-12.1
Transaction Summary
=====
Install 1 Package

Total download size: 3.4 M
Installed size: 18 M
Is this ok [y/N]:
```

```
Welcome to GNU Emacs, one component of the GNU/Linux operating system.
To follow a link, click Mouse-1 on it, or move to it and type RET.
To quit a partially entered command, type Control-g.

Important Help menu items:
Emacs Tutorial           Learn basic Emacs keystroke commands
Read the Emacs Manual    View the Emacs manual using Info
(Non)Warranty            GNU Emacs comes with ABSOLUTELY NO WARRANTY
Copying Conditions       Conditions for redistributing and changing Emacs
More Manuals / Ordering  Manuals How to order printed manuals from the FSF

Useful tasks:
Visit New File           Specify a new file's name, to edit the file
Open Home Directory      Open your home directory, to operate on its files
Customize Startup        Change initialization settings including this screen

GNU Emacs 29.4 (build 1, x86_64-redhat-linux-gnu) of 2025-06-17
Copyright (C) 2024 Free Software Foundation, Inc.
```

To install emacs make sure to install emacs-nox to get the command line version, as just writing emacs will download the GUI version. To install, run the command **sudo yum install emacs-nox**.

To run, type **emacs-nox**, and it should look like the second picture above. Emacs is a text editor that can be used to edit text and program. To exit, press ctrl-x ctrl-c.

<https://askubuntu.com/questions/23645/how-do-i-download-the-command-line-emacs-instead-of-the-gui-one>

```
ms3213@localhost:~$ sudo yum install fail2ban
Last metadata expiration check: 0:00:44 ago on Mon 17 Nov 2025 06:58:36 AM EST.
Dependencies resolved.
=====
Package                                Architecture      Version
=====
Installing:
fail2ban                               noarch             1.1.0-6.el10_0
Installing dependencies:
exim                                    x86_64             4.99-1.el10_2
fail2ban-firewalld                     noarch             1.1.0-6.el10_0
fail2ban-selinux                       noarch             1.1.0-6.el10_0
fail2ban-sendmail                     noarch             1.1.0-6.el10_0
fail2ban-server                       noarch             1.1.0-6.el10_0
libdb                                   x86_64             5.3.28-64.el10_0
libgsasl                               x86_64             1.10.0-12.el10_1
libidn                                  x86_64             1.42-4.el10_0
libns12                                x86_64             2.0.1-1.el10_0
libntlm                                 x86_64             1.8-1.el10_0
libopendmarc                           x86_64             1.4.2-33.el10_2
libspf2                                 x86_64             1.2.11-16.20210922
Installing weak dependencies:
publicsuffix-list                      noarch             20240107-5.el10
Transaction Summary
=====
Install 14 Packages

Total download size: 3.4 M
Installed size: 10 M
Is this ok [y/N]: y
```

To get fail2ban to install correctly on CentOS, you must first run **yum update -y** & **yum install epel-release -y**. **yum** update ensures everything is up to date. Epel stands for Extra Packages for Enterprise Linux, and running this command allows us to download other packages, including fail2ban. After these two steps, you can run **sudo yum install fail2ban** to get it installed, as seen in the above picture

```
ms3213@localhost:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:fail2ban(1)
ms3213@localhost:~$ sudo systemctl restart fail2ban
ms3213@localhost:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; disabled; preset: disabled)
   Active: active (running) since Mon 2025-11-17 08:28:23 EST; 4s ago
 Invocation: ed6af77890224a45b54a8273e6d64174
     Docs: man:fail2ban(1)
   Process: 8405 ExecStartPre=/bin/mkdir -p /run/fail2ban (code=exited, status=0/SUCCESS)
  Main PID: 8407 (fail2ban-server)
    Tasks: 3 (limit: 10548)
   Memory: 11.8M (peak: 12M)
      CPU: 333ms
   CGroup: /system.slice/fail2ban.service
           └─8407 /usr/bin/python3 -sP /usr/bin/fail2ban-server -xf start

Nov 17 08:28:23 localhost.localdomain systemd[1]: Starting fail2ban.service - Fail2Ban Service...
Nov 17 08:28:23 localhost.localdomain systemd[1]: Started fail2ban.service - Fail2Ban Service.
Nov 17 08:28:23 localhost.localdomain fail2ban-server[8407]: Server ready
```

Running **sudo systemctl status fail2ban** shows its status. If it is inactive, it may need to be restarted with **sudo systemctl restart fail2ban**.

```
ms3213@localhost:~$ sudo fail2ban-client status
[sudo] password for ms3213:
Status
|- Number of jail:      0
`- Jail list:
```

Running **sudo fail2ban-client status** will show you the status of the jails we have. In this case, we don't have any. If we wanted a jail for ssh we would have to enable one, as it is not there by default, like Ubuntu

<https://upcloud.com/resources/tutorials/install-fail2ban-centos/>

<https://github.com/fail2ban/fail2ban/discussions/3827>

<https://github.com/fail2ban/fail2ban/wiki/How-to-install-fail2ban-packages#centos--centos-stream--rhel-based-operating-systems>

<https://docs.fedoraproject.org/en-US/epel/>

<https://borneo.org/blog/2023/09/basic-fail2ban-commands/>

3.

```
ms3213@localhost:~$ sudo yum install cowsay
[sudo] password for ms3213:
CentOS Stream 10 - BaseOS
CentOS Stream 10 - BaseOS
CentOS Stream 10 - AppStream
CentOS Stream 10 - AppStream
CentOS Stream 10 - Extras packages
Dependencies resolved.
=====
Package                                Architecture
=====
Installing:
cowsay                                noarch

Transaction Summary
=====
Install 1 Package

Total download size: 45 k
Installed size: 80 k
Is this ok [y/N]:
```

```
ms3213@localhost:~$ cowsay "moooo"

  < moooo >
  -----
  \         ^ ^
   \      (oo)\_____.
    (__)\       )\/\
       ||----w |
       ||     ||

ms3213@localhost:~$
```

To install cowsay, run **sudo yum install cowsay** as shown in the first picture above. To use it, type **cowsay 'text here'** and it will print out the cow saying what you wrote. Show in the second picture.

```
ms3213@localhost:~$ sudo snap install lolcat
[sudo] password for ms3213:
2025-11-17T19:42:19-05:00 INFO Waiting for automatic snapd restart...
Download snap "lolcat" (1) from channel "stable"
lolcat 100.0.1 from moe (busyloop) installed
```

To install, run **sudo yum install snapd**, then **sudo systemctl enable --now snapd.socket**. After running **sudo ln -s /var/lib/snapd/snap /snap**. Restart the system, and you should now be able to install lolcat using **sudo snap install lolcat**

To run, type the command **echo**, then the text you want to change color, then lolcat to have it print out in rainbow. Example from picture above **echo "woooooooooow" lolcat**

4.

Install Vim using the command **sudo yum install vim-enhanced**. To run Vim, type **vim** “filename” and put the file you wish to use. Vim is a text editor that allows you to edit and search through text files without the need for a mouse. The above shows Vim being installed.

The **whereis** command can be used to find the location of each installation

```
tmux installed on 4/11/2026    /usr/bin/tmux
emacs installed on 4/11/2026   /usr/bin/emacs
fail2ban installed on 4/11/2026 /etc/fail2ban
cowsay installed on 4/11/2026  /usr/games/cowsay
lolcat installed on 4/11/2026  /var/lib/snapd/snap/bin/lolcat
vim updated on 4/11/2026       /usr/bin/vim
```